MASTER'S COMPREHENSIVE EXAMINATION

Part I: Theory (90 minutes) December 3rd, 2011

Instructions: Attempt <u>any three</u> out of the following four questions. Mention which three you would like to have graded. Open book and open notes.

1. Let X_1, X_2 be independent normal random variables with mean μ and variance 1.

   a) Find the mean squared error of $(X_1 + X_2)/2$.

   b) Find the mean squared error of $X_2 + 1$.

2. Let X_1, X_2, X_3 be independent random binomial variables with parameters $(1, p)$.

   a) Test p = ¼ vs. p = ½ at level .0156. That is, determine explicitly the rejection region of the most powerful test.

   b) Find the power of the test in (a).

3. Urn A has 3 white and 7 black balls. Urn B has 7 white and 3 black balls. We flip a fair coin. If the outcome is heads, then two balls from urn A are selected without replacement (that is, we do not place the first ball back), whereas if the outcome is tails, then two balls from urn B are selected without replacement. Suppose that the first selected ball is white, then:

   a) What is the probability that the coin landed heads

   b) What is the probability that the second selected ball is also white?

4. The joint density function of X and Y is given by

$$f(x; y) = \mathrm{x}\mathrm{e}^{-x(y+1)} \qquad x > 0; y.0:$$

   a) Find the conditional density of X, given $Y = y$, and that of Y, given $X = x$.

   a) Find the density function of $Z = XY$.